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Electrical and Computer Engineering

Department Computer Networks Laboratory

ENCS4130

Report 2

Experiment No. 9

Cisco ASA Firewall Configuration

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Section: 3

BIRZEIT

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Abstract

The purpose of this experiment is to understand security zone architecture and traffic control mechanisms through Cisco ASA firewall configuration. We examine security level hierarchies while implementing NAT (selecting static mapping for DMZ services and dynamic PAT for internal clients), access control policies, and DMZ deployment. The lab establishes a multi-tier network (internal, DMZ, external) with interface security levels, integrates OSPF/BGP routing between edge routers, and deploys WPA2-secured wireless access. Validation through DNS resolution, HTTP/HTTPS access, and ICMP tests demonstrates stateful inspection's role in permitting legitimate traffic while blocking unauthorized attempts, offering practical insights into enterprise security design decisions.

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Acronyms and Abbreviations

ASA: Adaptive Security Appliance

NAT: Network Address Translation

PAT: Port Address Translation

DMZ: Demilitarized Zone

ACL: Access Control List

OSPF: Open Shortest Path First

BGP: Border Gateway Protocol

AS: Autonomous System

DNS: Domain Name System

ICMP: Internet Control Message Protocol

WPA2: Wi-Fi Protected Access 2

PSK: Pre-Shared Key

AES: Advanced Encryption Standard

TCP: Transmission Control Protocol

DHCP: Dynamic Host Configuration Protocol

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1. Introduction

1.1. Network Security and Firewalls

A firewall is a network security device or software that monitors and controls incoming and outgoing network traffic based on predetermined security rules. Firewalls serve as the first line of defense in network security, acting as a barrier between trusted internal networks and untrusted external networks, such as the Internet. Their primary role is to block unauthorized access while permitting legitimate communication, thereby protecting sensitive data and resources from cyber threats. There are several types of firewalls, each suited for different security requirements:

- **Packet filtering firewalls** examine packets individually and allow or block them based on IP addresses, port numbers, and protocols.

- **Stateful inspection firewalls** track the state of active connections and make decisions based on the context of the traffic.

- **Proxy firewalls** function as intermediaries, forwarding requests and responses between users and external networks, which helps hide internal network details.

- **Next-generation firewalls (NGFWs)** combine traditional firewall functions with advanced features like intrusion prevention, application awareness, and deep packet inspection.

- **Cloud firewalls** are hosted in the cloud and provide security for cloud-based infrastructure and services.

By establishing a secure perimeter, firewalls are crucial for preventing unauthorized access, cyberattacks, and data breaches, ensuring the integrity and confidentiality of modern networked environments [1].

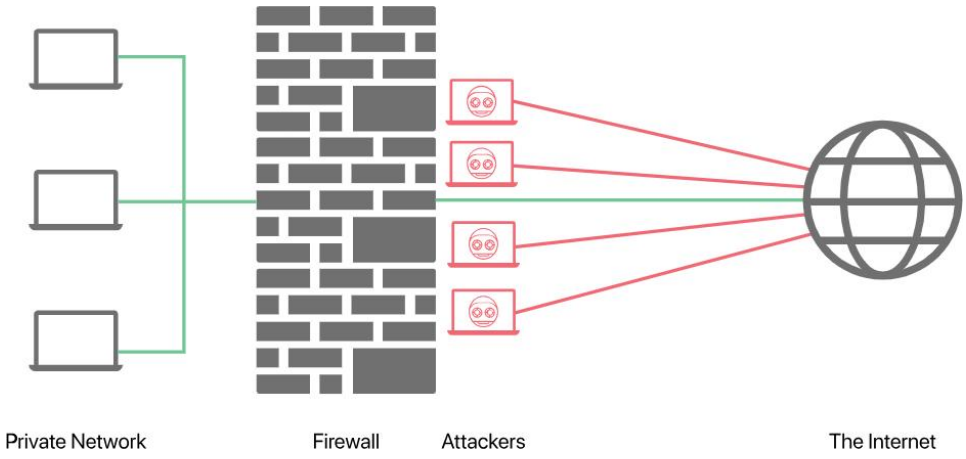

Figure 1: Firewall protects the private network from Internet threats.

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1.2. Cisco Adaptive Security Appliance (ASA)

The Cisco Adaptive Security Appliance (ASA) is a powerful, multifunctional security device widely used to protect modern enterprise networks. As a comprehensive security solution, Cisco ASA integrates several essential features into a single platform, including a robust firewall, Network Address Translation (NAT), Virtual Private Network (VPN) support, intrusion prevention, antivirus capabilities, and stateful packet inspection. By default, the ASA operates with a "default-deny" security policy, meaning all inbound traffic from less trusted networks is blocked unless explicitly permitted. Security administrators can customize and fine-tune traffic flow using Access Control Lists (ACLs), which define specific rules to allow or deny different types of network traffic. This flexibility, combined with advanced inspection and protection mechanisms, makes Cisco ASA a leading choice for organizations seeking to secure their networks against a wide range of threats [2].

1.3. Demilitarized Zone (DMZ)

A Demilitarized Zone (DMZ) is a physical or logical subnetwork that separates an organization's internal, private network from untrusted external networks, such as the Internet. The main purpose of a DMZ is to host public-facing services—such as web servers, mail servers, and DNS servers—in a zone that is isolated from the internal network. This setup provides an additional layer of security: if an attacker compromises a system in the DMZ, the internal network remains protected. By segmenting these services, organizations can tightly control the flow of traffic between the DMZ, the Internet, and the internal network, reducing the risk of unauthorized access to sensitive internal resources [3].

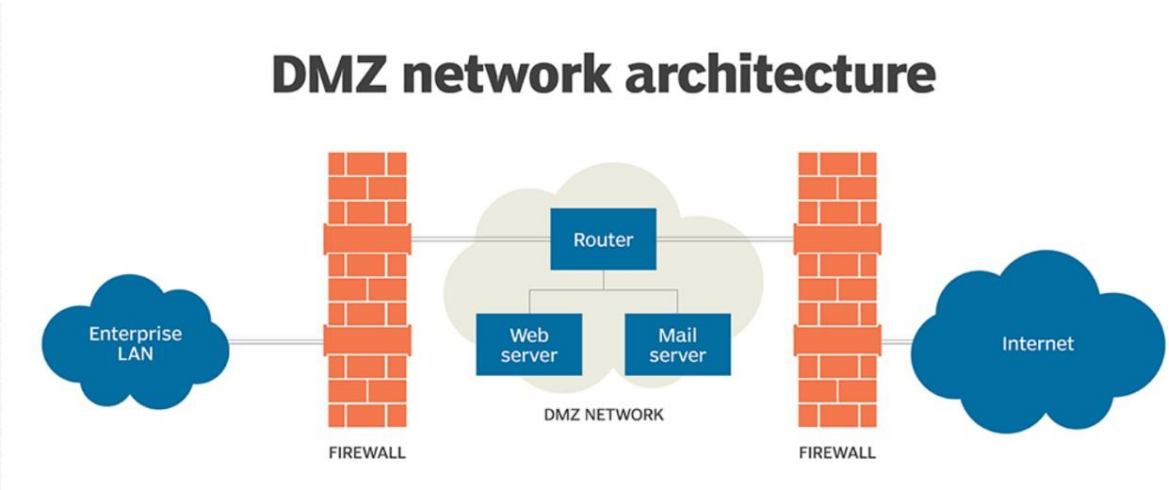


Figure 2: DMZ network architecture

1.4. Network Address Translation (NAT) and PAT

Network Address Translation (NAT) is a technique used to translate private (internal) IP addresses into public (external) IP addresses, allowing devices within a private network to access external networks such as the Internet. NAT provides both security and address conservation, as private addresses are hidden from external users. There are several types of NAT:

• Static NAT maps a single private IP address to a single public IP address, typically used for servers that must be reachable from external networks.

• Dynamic NAT maps private addresses to public addresses from a pool, assigning them on a first-come, first-served basis.

• Port Address Translation (PAT), also known as NAT overload, maps multiple private IP addresses to a single public IP address by using different port numbers for each session. PAT is the most common form of NAT, enabling efficient use of public IPs [4].

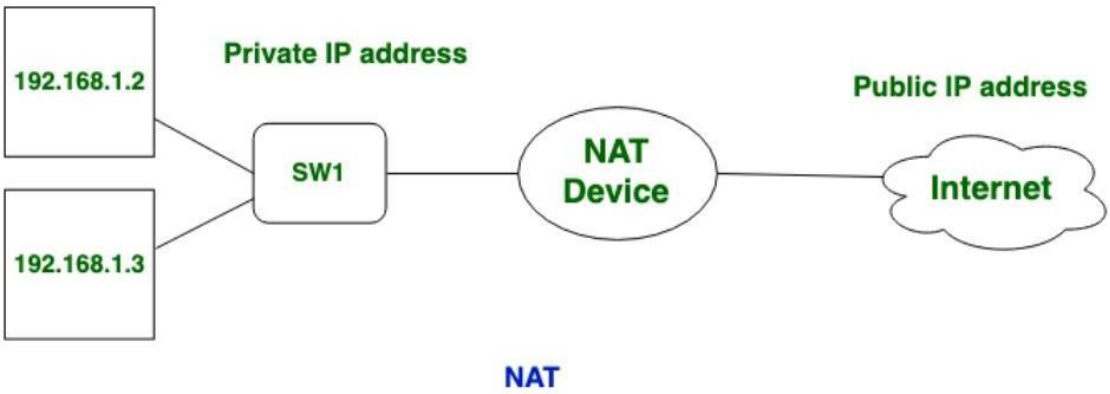
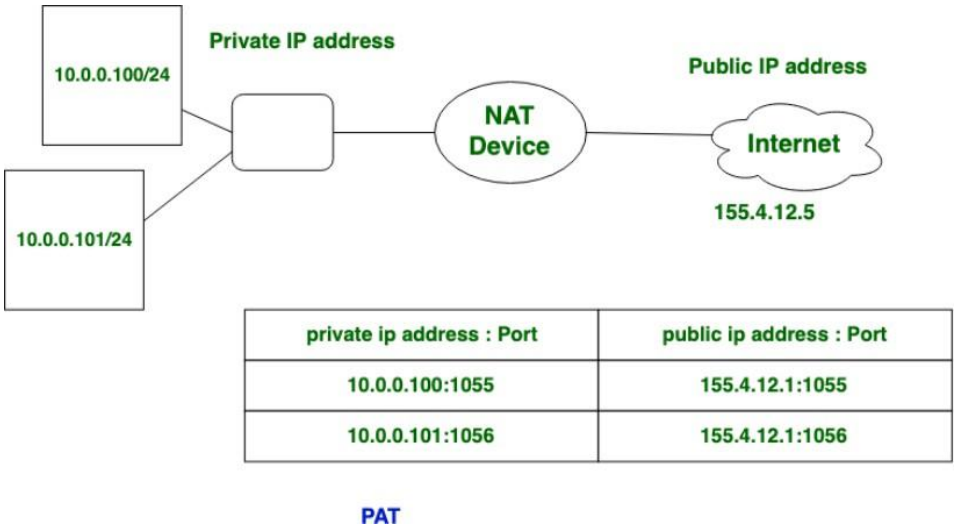


Figure 3: Network Address Translation (NAT)



private ip address : Port	public ip address : Port
10.0.0.100:1055	155.4.12.1:1055
10.0.0.101:1056	155.4.12.1:1056

Figure 4: Port Address Translation(PAT)

1.5. Dynamic Host Configuration Protocol (DHCP)

DHCP is a network protocol that automates the assignment of IP addresses and network parameters (DNS, subnet mask, default gateway) to devices. This simplifies network management, reduces errors, and optimizes IP usage. DHCP operates through a four-step process: Discover, Offer, Request, and Acknowledgment (ACK), enabling clients to communicate on the network with minimal manual intervention[1][2].

1.6. Domain Name System (DNS)

DNS translates human-readable domain names into IP addresses, enabling devices to locate web servers. When a user types a URL, the system first checks the local cache. If unresolved, it forwards a recursive or iterative query to DNS servers, progressing from root to TLD to authoritative servers until the correct IP is found. This allows the browser to connect and display the requested web page[1][3].

1.7. Web Server

A Web Server stores and delivers web content using HTTP protocols. When a user enters a URL, DNS resolves the domain to an IP address, and a TCP connection is established via a three-way handshake. The browser sends an HTTP request, the server responds with the content, and the browser renders the web page for viewing[1][4].

1.8. Email Server

An Email Server manages sending, receiving, and storing emails using protocols like SMTP for outgoing and POP3/IMAP for incoming messages. It uses DNS and MX records to route emails and ensures secure communication through authentication and encryption. These features provide reliable email delivery across networks[1].

1.9. Open Shortest Path First (OSPF) OSPF

is a scalable, efficient link-state routing protocol based on the SPF algorithm. It supports network segmentation into areas, reduces routing overhead, and improves scalability by sharing only link-state changes. OSPF uses IP multicast for routing updates, supports VLSM, and enhances security with packet authentication, ensuring reliable and secure data transmission[1].

1.10. Border Gateway Protocol (BGP)

BGP (Border Gateway Protocol) is the routing protocol used to exchange routing information between different autonomous systems (AS) on the internet. It determines the best path for data to travel by considering factors such as network policies, rules, and paths, rather than simply the shortest distance. BGP is essential for maintaining internet stability and avoiding routing loops or traffic congestion. It can be used for load balancing and failover, ensuring resilient and efficient communication between large networks[8].

1.11. Access List (ACL)

Access Control Lists (ACLs) are sets of rules used to filter network traffic and enhance security by controlling which packets are allowed or denied based on criteria such as IP addresses, protocols, and port numbers. ACLs can be used to control access between network segments, limit routing updates, and enforce network policies. In this experiment, ACLs were applied to the ASA firewall to permit specific types of traffic—such as HTTP and DNS—while blocking unauthorized access to sensitive resources. By enforcing precise traffic control policies, ACLs contribute to a secure and efficient network environment[1].

2. Procedure and Discussion

2.1 Building the topology

I created the topology using the 5506-X device for the firewall, Router-PT for the routers, and Switch-PT for the switches. For the end devices, I used PC-PT, Laptop-PT, and other available options. To connect all devices, I utilized the "Automatically Choose Connection Type" option to make the linking process easier.

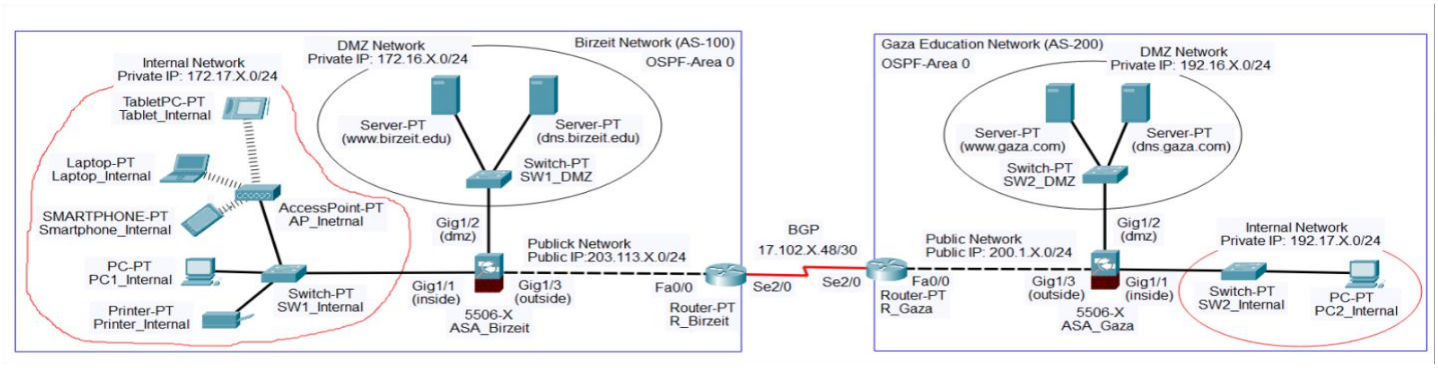


Figure 5: Firewall Topology

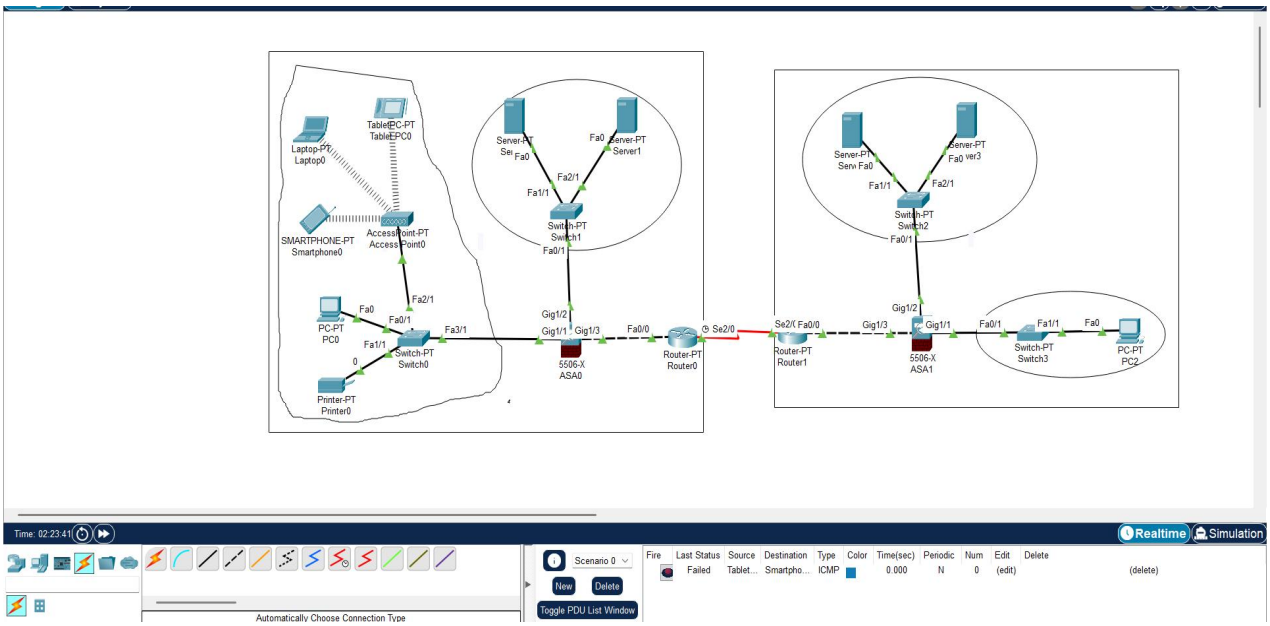


Figure 6: My Topology

2.2 Routers Configuration

2.2.1. Configuring Router0 Interfaces

I configured **Router R0 (R_Birzeit)** interfaces with the following commands:

(R_Birzeit)> enable

(R_Birzeit)# configure terminal

(R_Birzeit)(config)# interface FastEthernet0 /0

(R_Birzeit)(config - if)# ip address 203.113.74.2 255.255.255.0

(R_Birzeit) (config - if)# no shutdown

(R_Birzeit) (config)# interface Serial2 /0

(R_Birzeit)(config - if)# ip address 17.102.74.49 255.255.255.252

(R_Birzeit)(config - if)# no shutdown

These commands were executed to configure Router0's (R_Birzeit) interfaces. The same configuration steps were carried out on Router1 (R_Gaza) with different IP addresses.

2.3 Configuring OSPF and BGP Routing Protocols

In this step, I configured the routing protocols OSPF and BGP on both routers to enable dynamic routing between the networks.

On Router0 (R_Birzeit)

I configured OSPF with process ID 1 to advertise the public network and redistributed BGP routes into OSPF:

(R_Birzeit)(config)# router ospf 1 Router0

(R_Birzeit)(config-router)# network 203.113.74.0 0.0.0.255 area 0 Router0

(R_Birzeit)(config-router)# redistribute bgp 100 subnets

Then, I configured BGP by specifying the external BGP (eBGP) neighbor Router1(R_Gaza) and redistributing OSPF routes into BGP:

(R_Birzeit)(config)# router bgp 100

(R_Birzeit)(config-router)# neighbor 17.102.74.50 remote-as 200

(R_Birzeit)(config-router)# redistribute ospf 1

On Router1(R_Gaza)

I repeated the same steps on Router1, adjusting the internal network IP range, autonomous system (AS) number, and neighbor IP accordingly.

2.4 Servers Configuration

2.4.1. Birzeit Web Server

- Assigned a static IP address to the Birzeit Web Server as shown in Figure 2.
- Enabled HTTP and HTTPS (secure HTTP) protocols to allow web access.
- Customized the index.html page to display the title: **"Birzeit University"**.

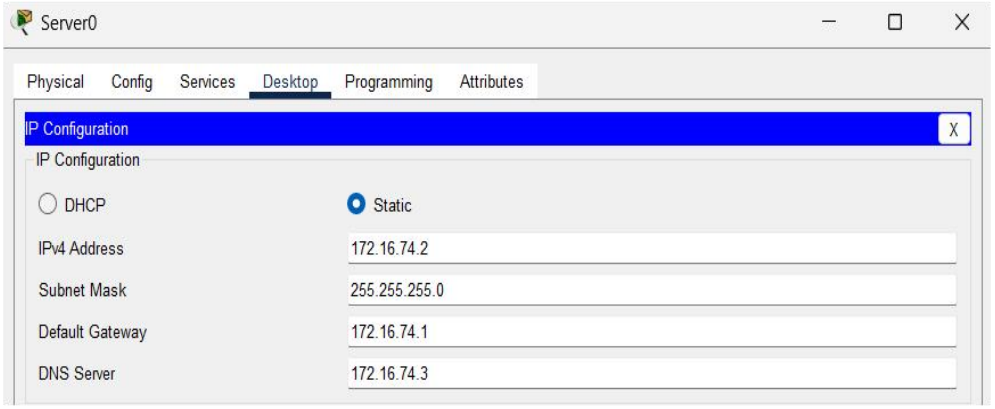


Figure 7 : IP configuration of server0 (Birzeit Web server)

2.4.2. Birzeit DNS Server (dns.birzeit.edu)

I assigned a static IP configuration to the Birzeit DNS server. Only the DNS service was enabled on this server to manage domain name resolutions.

The following resource records were added to the DNS server:

- The domain www.birzeit.edu points to the IP address of the Birzeit web server (Type A record).
- The domain **dns.gaza.com** points to the IP address of the Gaza DNS server (Type A record).
- The domain **gaza.com** uses **dns.gaza.com** as its name server (Type NS record).

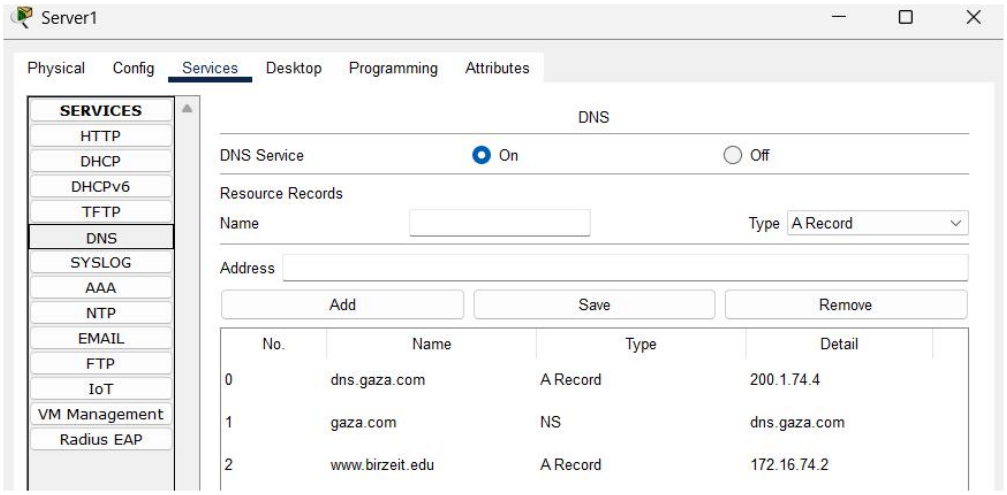


Figure 8: DNS service configuration of Birzeit DNS server.

2.4.3. Gaza Web Server

I assigned a static IP address to the Gaza Web Server and enabled both HTTP and HTTPS protocols to allow secure web access.

I also customized the server's homepage by editing the index.html file to display the title ““Gaza Education Net-work”

2.4.4 Gaza DNS Server (dns.gaza.com)

I assigned a static IP configuration to the Gaza DNS Server. Only the DNS service was enabled on this server to manage domain name resolutions.

The following resource records were added to the Gaza DNS server

- The domain www.gaza.com points to the IP address of the Gaza web server (Type A record).
- The domain **dns.birzeit.edu** points to the public IP address of the Birzeit DNS server (Type A record).
- The domain **birzeit.edu** uses **dns.birzeit.edu** as its name server (Type NS record).

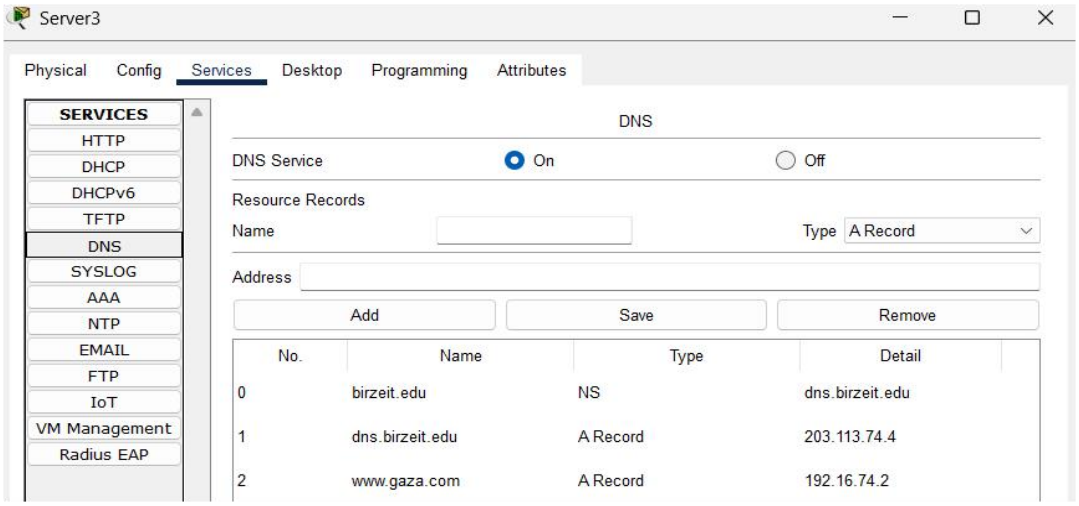


Figure 9: DNS service configuration of Gaza DNS server.

2.5 ASA Firewall Configuration

2.5.1 Configuring Privileged Mode Password

The default password for the privileged mode on the ASA firewall is blank. Therefore, when prompted for the privileged mode password, just press the **Enter** key.

The following commands were used to change the password to **encs4130**:

```
ASA-Birzeit> enable Password:
ASA-Birzeit# configure terminal
ASA-Birzeit(config)# enable password encs4130
```

2.5.2 Configuring Interfaces and Security Zones

Each ASA interface must be configured with a name, security level, IP address, and enabled.

The recommended security levels and roles for the Birzeit network are as follows:

- **Inside (Sec Level 100):** Represents the most trusted zone, which is Birzeit's internal network.
- **DMZ (Sec Level 50):** A semi-trusted zone that hosts externally accessible servers.
- **Outside (Sec Level 0):** The untrusted zone connected to the public Internet.

To configure the inside interface (internal network):

```
ASA-Birzeit(config)# interface Gig1/1
ASA-Birzeit(config-if)# nameif inside
ASA-Birzeit(config-if)# security-level 100
ASA-Birzeit(config-if)# ip address 172.17.74.1 255.255.255.0
ASA-Birzeit(config-if)# no shutdown
```

Repeat the same steps to configure the outside and DMZ interfaces using the appropriate interface names, security levels, and IP addresses.

2.5.3 Configuring PAT for the Internal Network

PAT (Port Address Translation) enables the devices in the Birzeit internal network to share a single public IP address when accessing the Internet.

To map all internal devices in the 172.17.774.0/24 subnet to the ASA firewall's outside interface IP address, I used the following configuration:

```
ASA-Birzeit(config)# object network INTERNAL-PAT
ASA-Birzeit(config-network-object)# subnet 172.17.74.0 255.255.255.0
ASA-Birzeit(config-network-object)# nat (inside,outside) dynamic interface
```

This configuration translates the private IP addresses of all devices inside the network to the public IP of the outside interface, allowing secure Internet access.

2.5.4. Configuring Static NAT for the DMZ Network

To allow external access to DMZ servers, static NAT is configured to map each private IP to a public IP from the 203.113.74.0/24 subnet. The ASA outside interface uses 203.113.74.1, and the R Birzeit router uses 203.113.74.2, so public IP assignments start from 203.113.74.3. Accordingly, the Web server with private IP 172.16.74.2 is statically mapped to the public IP 203.113.74.3.

```
ASA_Birzeit ( config )# object network DMZ - WEB - NAT
ASA_Birzeit ( config - network - object )# host 172.16.74 .2
ASA_Birzeit ( config - network - object )# nat ( dmz , outside ) static 203.113.74.3
```

2.5.5. Enabling External Access to DMZ Servers

To allow external users to access the Web and DNS services hosted in the DMZ, I created an Access Control List (ACL) that permits the required traffic from external networks to the DMZ servers.
The following ACL rules were configured:

```
ASA-Birzeit(config)# access-list OUTSIDE-ACCESS extended
permit tcp any host 172.16.74.2 eq www
ASA-Birzeit(config)# access-list OUTSIDE-ACCESS extended
permit udp any host 172.16.74.3 eq domain
```

Then, I applied the **OUTSIDE-ACCESS** ACL to the outside interface:

```
ASA-Birzeit(config)# access-group OUTSIDE-ACCESS in interface outside
```

2.5.6. Enabling Return Traffic for HTTP, DNS, and ICMP to the Internal LAN

The ASA firewall is stateful, meaning it can track connections and allow return traffic for inspected protocols. This prevents legitimate responses from the DMZ or the Internet to internal clients from being blocked.
By default, **DNS inspection is enabled** on the ASA. To allow **HTTP responses**, HTTP inspection must be explicitly enabled by updating the global policy map using the following commands:

```
ASA-Birzeit(config)# policy-map global_policy
ASA-Birzeit(config-pmap)# class inspection_default
ASA-Birzeit(config-pmap-c)# inspect http
```

2.5.7 Configuring Default Routing

To enable Internet access for internal and DMZ devices, I configured a default route that forwards all external traffic to the Router0.

The following command was used:

ASA-Birzeit(config)# route outside 0.0.0.0 0.0.0.0 203.113.74.2

2.5.8 Configuring ASA as a DHCP Server for the Internal LAN

To enable dynamic IP address allocation for internal LAN devices, I configured the ASA to act as a DHCP server on the inside interface.

Since the IP address 172.17.74.1 is reserved for the inside interface and the printer requires a static IP configuration , I defined a DHCP address pool of 100 addresses starting from 172.17.74.3 to 172.17.74.102.

I also specified the DNS server in the DHCP configuration.

The following commands were used:

ASA-Birzeit(config)# dhcpd address 172.17.74.3-172.17.74.102 inside

ASA-Birzeit(config)# dhcpd dns 172.16.74.3

ASA-Birzeit(config)# dhcpd enable inside

2.6 Access Point Configuration

The access point provides wireless connectivity within the internal network. To configure the wireless network, I went to the **Config** tab of the access point and set the following parameters:

- **Name:** BZU-WiFi
- **Security Model:** WPA2-PSK
- **Pre-Shared Key (PSK):** EnCs4130
- **Encryption Type:** AES

Wireless devices must enter the pre-shared key to connect to the access point.

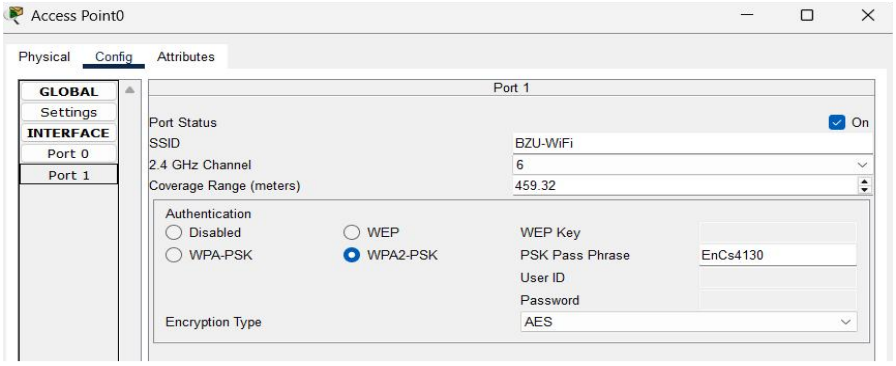


Figure 10: Configuring the access point for Birzeit internal network.

2.7 Connecting Wireless Devices and Configuring Dynamic IP Addresses

The tablet, laptop, and smartphone connect wirelessly to the internal network through the access point. However, unlike the tablet and smartphone, the laptop does not have a built-in wireless interface. To enable wireless connectivity, I replaced the laptop's wired LAN module with a **WPC300N wireless adapter**.

The steps to install the wireless module on the laptop are:

1. Click on the laptop and navigate to the **Physical** tab.
2. Turn off the device and remove the existing FastEthernet module.
3. Insert the **WPC300N wireless adapter** from the module list.
4. Power the laptop back on to activate the new wireless module.

Figure 11 illustrates how the laptop connects to the **BZU-WiFi** network and obtains an IP address dynamically using DHCP.

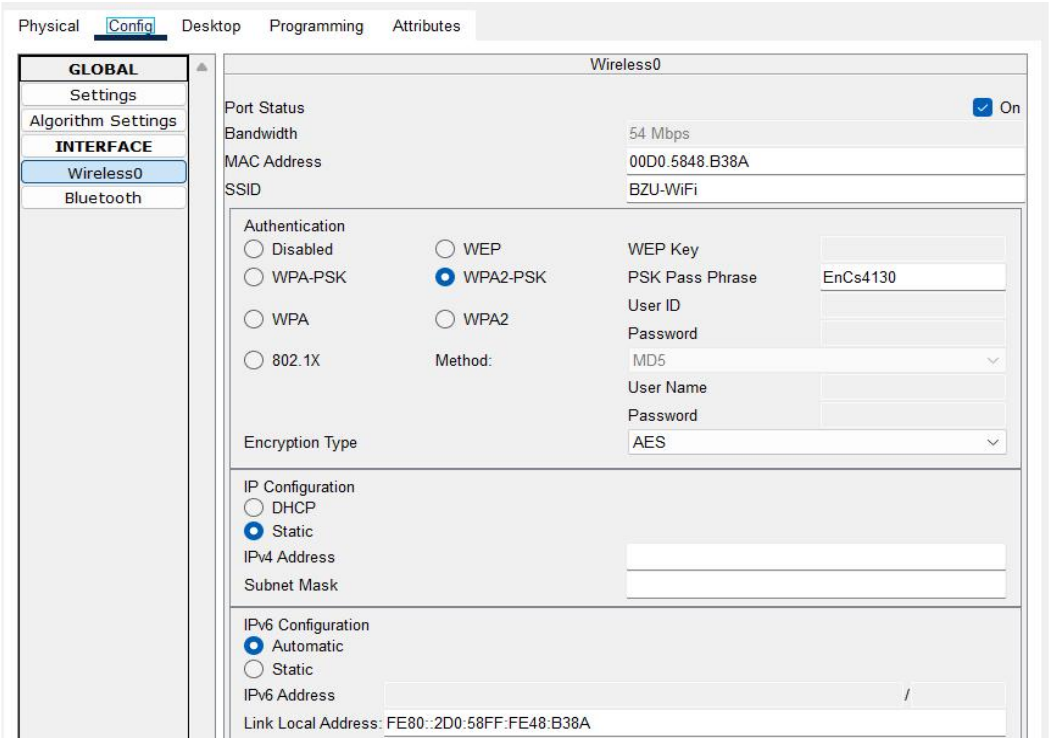


Figure 11: Connecting the laptop to BZU-WiFi and enabling dynamic IP configuration.

2.8 Configuring End Devices with Static IP Addresses

The **Printer_Internal** and **PC_Gaza** were configured with static IP addresses, including their default gateway and DNS server settings. Each device must have a unique IP address within the valid host range of its subnet.

Figure 9 shows the configuration of the Printer.

Next, the **PC_Gaza** was configured to use a static IP address.

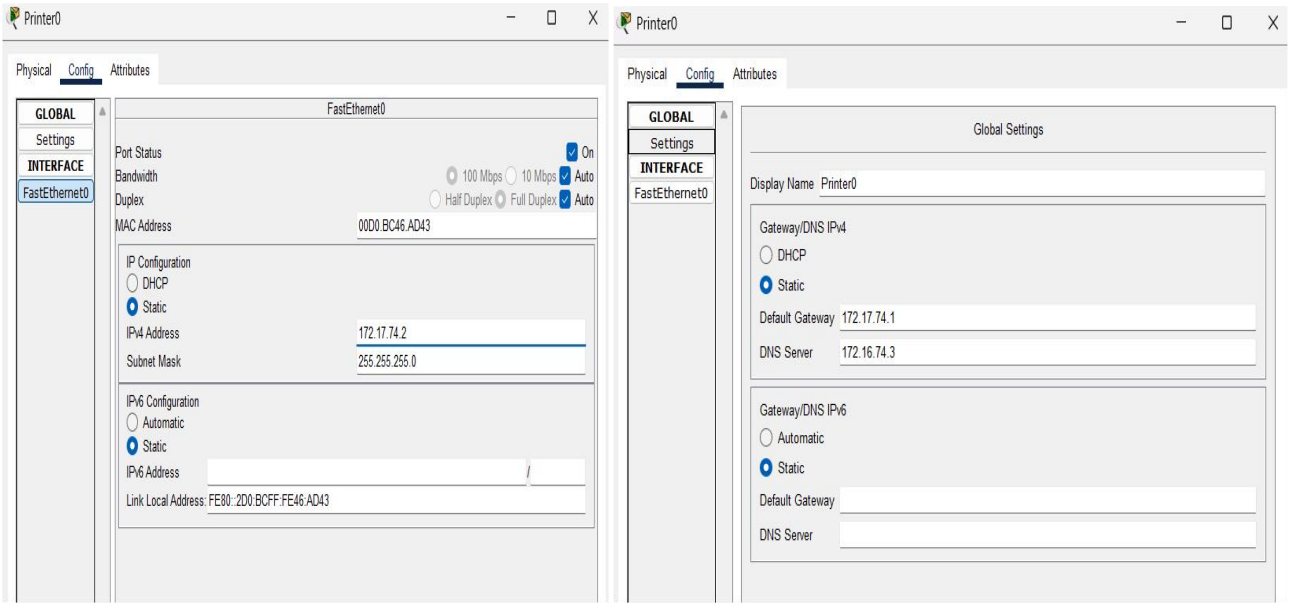


Figure 12: Configuring the Printer

3. Testing Connectivity

Test 1 – Pinging the R_Gaza from Laptop Internal (Laptop0):

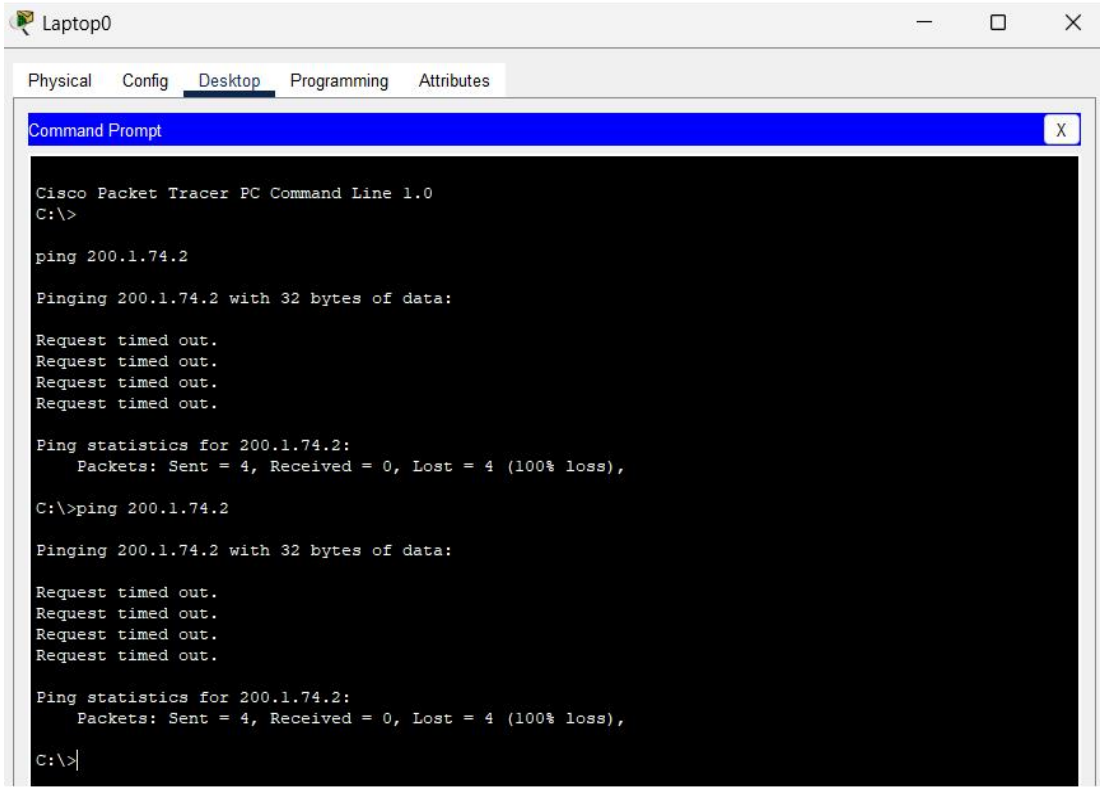


Figure 13: Blocked attempt from Laptop0 to R_Gaza.

I tried to send ping requests from Laptop Internal (Laptop 0) to a R_Gaza, but the requests failed, as you can see in the screenshot. This happened because the ASA firewall blocks any traffic coming from the Laptop Internal.

The ping commands sent 2 packets each time, but none got a reply, which means 100% packet loss. This shows that the firewall is doing its job by blocking these ping attempts from outside the network.

Test 2 – Attempting to Ping a Host in Birzeit Network from PC2 Internal:

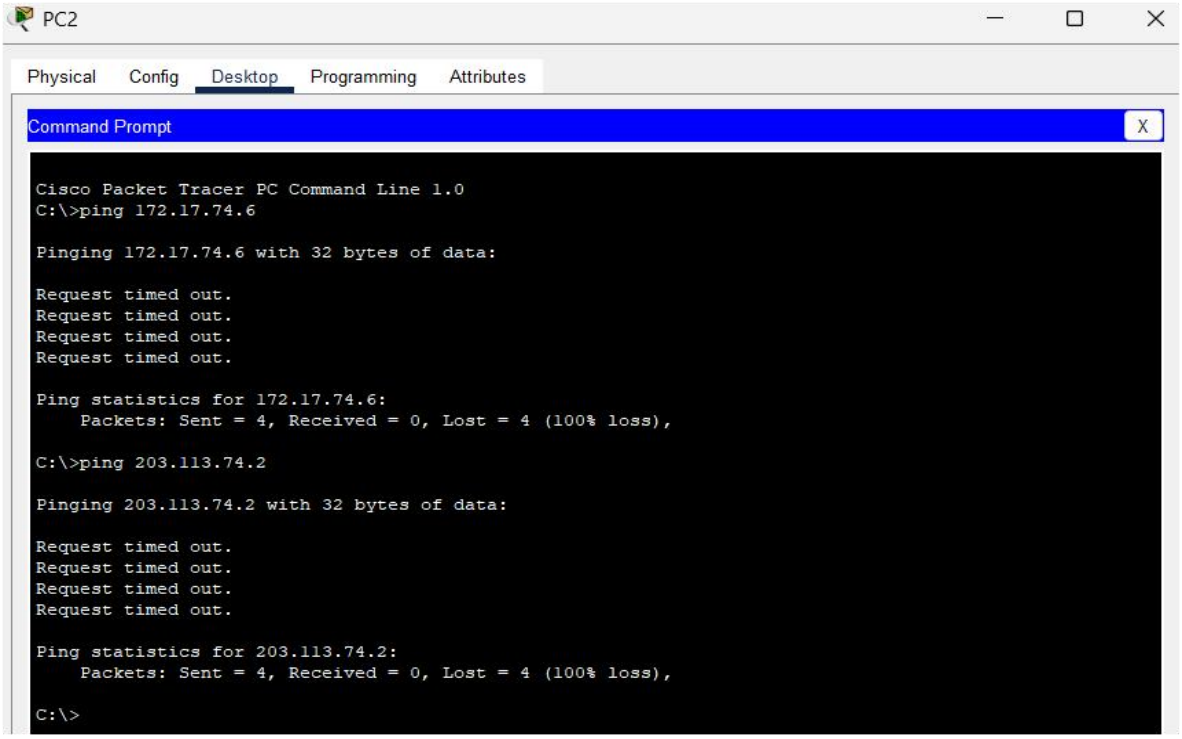


Figure 14: Blocked attempt from PC2 to Host in Birzeit Network.

I also tried to ping the Host in Birzeit Network from PC2, but this test failed as well. As shown in the attached screenshot, all ping requests timed out with no replies, resulting again in 100% packet loss. This indicates that the target device is either unreachable or that there is a firewall or network setting blocking the ping responses.

Test 3 – Accessing “www.gaza.com” from PC2 Internal:

To verify the end-to-end functionality of DNS resolution and HTTP access between the PC2 , the DMZ, and the Gaza internal network, the following test was performed:

Test1 – Accessing “www.gaza.com” from PC2 :

From the PC2 , an attempt was made to access the DMZ web server using the domain name www.gaza.com.

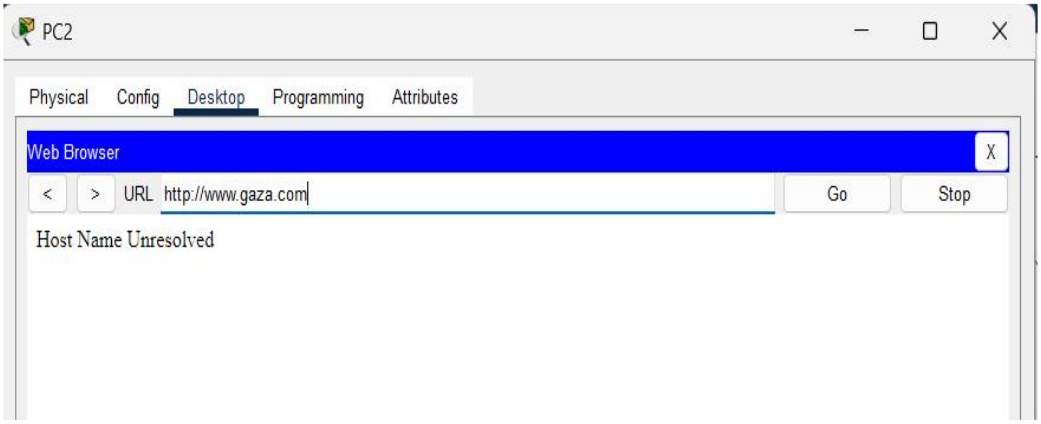


Figure 15: Try to Access “www.gaza.com” from PC2 Internal.

As shown in Figure 15, the web browser displayed the error message “Host Name Unresolved”, indicating a failure in DNS resolution. This means the PC was unable to translate the domain name to an IP address, and therefore the connection to the web server was unsuccessful.

Test 4 – Accessing “www.birzeit.edu” from PC2 Internal:

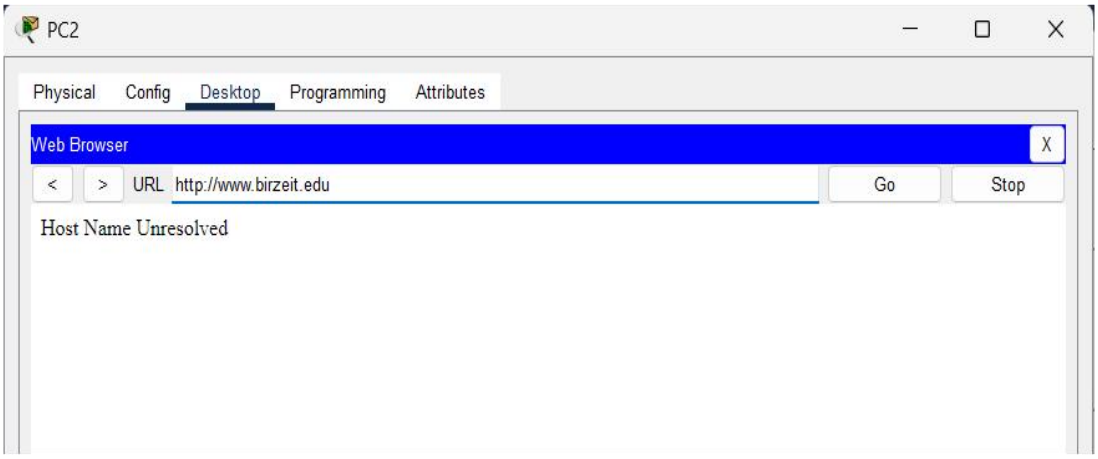


Figure 16: TRY TO ACCESS BIRZEIT PAGE .

As shown in Figure 16, the web browser displayed the error message “Host Name Unresolved”, indicating a failure in DNS resolution. This means the PC2 was unable to translate the domain name to an IP address, resulting in unsuccessful connection to the web server.

This suggests that the DNS service or network configuration for the internal wireless network is not correctly set up or there is a connectivity issue blocking DNS resolution.

4. Conclusion

In this experiment, I configured the Cisco ASA firewall to protect the network. I learned how to set IP addresses, create security zones, and use NAT to share one public IP. I also set up access rules to allow traffic to DMZ servers and inspected important protocols like HTTP and DNS.

The tests showed some connection problems, which helped me understand the importance of correct configuration and troubleshooting. Overall, this lab improved my knowledge of firewall basics and network security, while enhancing my understanding of real-world network setup, the significance of DHCP, and the impact of security configurations on inter-network communication.

5. Feedback

This experiment was one of the most challenging and time-consuming in the lab due to the complexity of configuring the ASA firewall, integrating multiple routing protocols, and ensuring secure access between network zones. Setting up NAT, ACLs, and DHCP on the ASA, as well as verifying communication between the internal network, DMZ, and external networks, required close attention to detail and thorough testing. Despite encountering some issues with DNS resolution and ICMP testing due to limitations in Packet Tracer, the overall configuration process was smooth, and I was able to apply practical troubleshooting techniques to identify and resolve misconfigurations. This experiment was highly valuable as it deepened my understanding of how firewalls, routing protocols, and network services interact in a real-world environment. It also improved my skills in configuring and managing security policies on enterprise-grade network devices.

6. References

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